UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #6

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: ________________________________
Due Date: Saturday April 11th by 12:00 noon

Number of points.........................100
Your score..............................

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Scan your Assignment and create a PDF file
- PDF File Name: Assignment6_XY.PDF
  - X: First letter of your first name
  - Y: Your last name
  - Example: Assignment6_MCaracotsios.PDF
- Email your PDF file to: Suzane Carneiro svieir2@uic.edu by the due date (Saturday April 11) and time (12:00 noon)

Note: All problems are worth 100 points. You final score will be the average value

Problem 6A: A hot hydrocarbon feed stream consisting of propane(1), toluene(2), ethyl benzene(3) and p-xylene(4) enters a shell and tube condenser at temperature $T = 275\,^{o}C$ and pressure $P = 10\,bar$ with an overall molar composition $[z_1, z_2, z_3, z_4] = [0.33, 0.30, 0.15, 0.22]$ and mass flow rate $F = 150,000\,kg/hr$. The hydrocarbon mixture is then cooled under constant pressure until a liquid stream of volumetric flow rate $L = 700\,gpm$ is produced. (See process flowsheet diagram below). The liquid is then pumped to an inventory tank for sales. The discharge pressure of the pump is set at $P = 20\,bar$

1. Calculate the bubble point and dew point temperature of the hydrocarbon feed mixture.
2. Calculate the temperature and molar composition of the liquid and vapor streams after the cooling process.
3. Calculate the amount of heat in <u>*kW*</u> removed from the feed stream during the cooling process.
4. Calculate the power requirements in <u>*kW*</u> for the pump if its efficiency is 75%
5. Calculate the temperature rise across the pump.

For all calculations assume that both the vapor and liquid phases behave ideally and that Raoult's law applies for vapor liquid equilibrium calculations for all components except propane. Propane is at supercritical conditions so you must use Henry's law with coefficient:

$$H_1[bar] = \exp\left(9.13237 - \frac{1940.2}{T}\right) \quad T[=]K$$